

ORDINARY LEAST SQUARES (OLS) REGRESSION

DATA 3100

Prof. Bell

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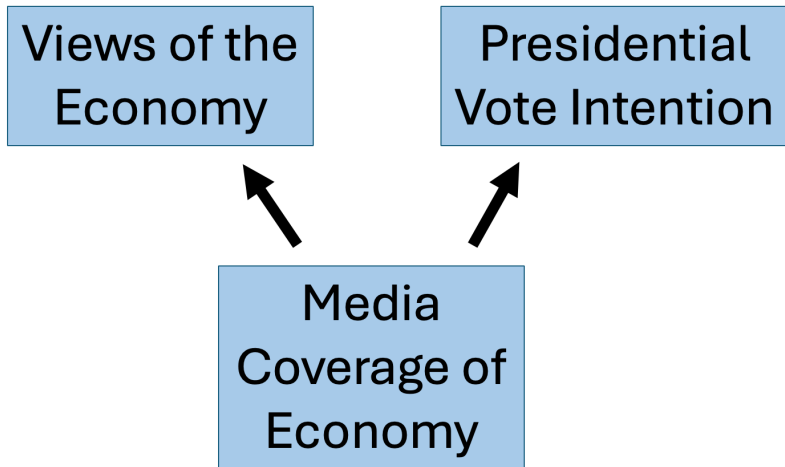
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Confounder

A variable that explains change in both the independent and dependent variable.

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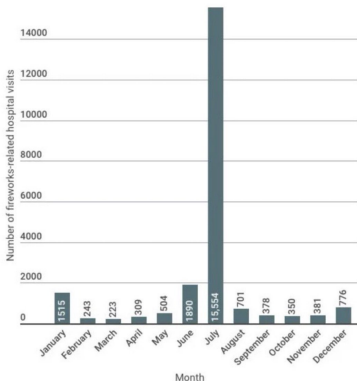


WHAT IS REGRESSION?

"It's most dangerous to launch fireworks in July."

IV: Time

DV: Fireworks-related hospitalizations

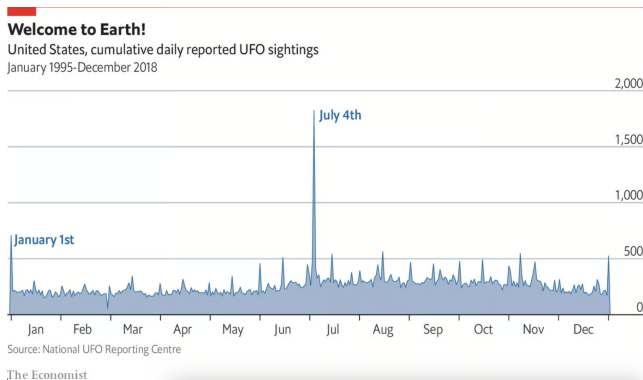


WHAT IS REGRESSION?

"If aliens exist, they will arrive on July 4th."

IV: Time

DV: UFO Sightings



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 - ▶ What if we have two continuous variables? Income, age, and years of education are common variables that we may not want to force into two discrete categories.
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Confounder

A variable that explains change in both the independent and dependent variable.

- ▶ When we fail to account for confounders, we face **omitted variable bias** and our estimates are not accurate.

ESTIMATING THE REGRESSION (BEST FIT) LINE

The linear regression equation is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Y = dependent variable

β_0 = intercept

β_1 = slope, also called a coefficient

X = independent variable

ϵ = error

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EXAMPLE

You want to estimate the effect of income on the allocation to welfare applicants. The linear regression equation is:

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If “Income” in thousands has a value of 100, and the intercept is \$605, and the coefficient is -.5, what is our estimated mean of “Allocation”?

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$$\text{Allocation} = 555$$

So the coefficient (β_1) is the effect of a **one-unit** change in income (thousands) on the mean allocation.

WHAT IS ERROR?

Recall that the linear regression equation is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Y = dependent variable

β_0 = intercept

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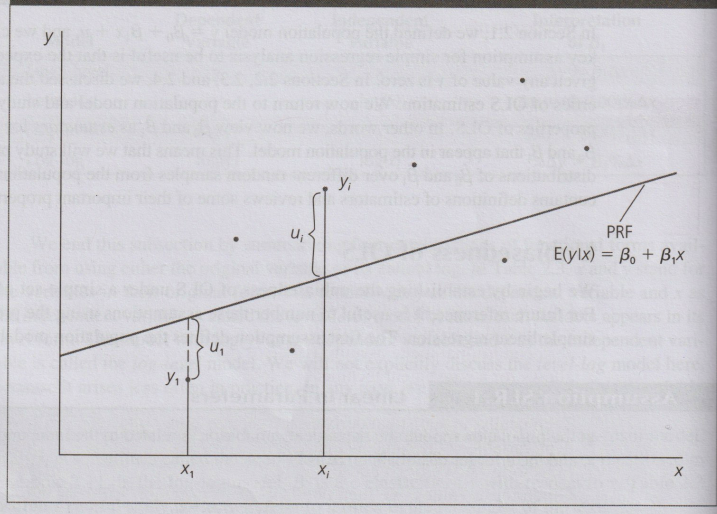
ϵ = error

WHAT IS ERROR?

- Error (ϵ or u) is also called the residual (left over from $Y = \beta_0 + \beta_1 X$, our best fit line)

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FIGURE 2.7 Graph of $y_i = \beta_0 + \beta_1 x_i + u_i$.



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- Error (ϵ or u) is also called the residual (left over from $Y = \beta_0 + \beta_1 X$, our best fit line)
- Our goal in regression is to fit the best line that minimizes the error
- However, we can never get the $\epsilon = 0$, and often we don't even get close. We just do the best we can to make the “best” best fit line.

MULTIPLE LINEAR REGRESSION

The multiple linear regression equation is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Y = dependent variable

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What is the interpretation of β_1 ?

β_1 is the effect of a one-unit change in X_1 on the mean of Y , holding X_2 constant (the independent effect of X_1 on Y)

MULTIPLE LINEAR REGRESSION

The multiple linear regression equation is:

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What is the interpretation of β_2 ?

β_2 is the effect of a one-unit change in X_2 on the mean of Y , holding X_1 constant (the independent effect of X_2 on Y)

NON-CONTINUOUS DEPENDENT VARIABLES

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- What about categorical dependent variables, like party ID? The linear regression model is not well suited for these.

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- One of the assumptions of linear regression is that the relationship between X and Y is linear (you can draw a best fit line)
- This assumption is usually fine when we are working with continuous dependent variables like income
- What about categorical dependent variables, like party ID? The linear regression model is not well suited for these.
- What about binary dependent variables, like support for a policy?

We call this a linear probability model because it estimates the effect of a one-unit change in X on the *probability* (percent chance) of a value of “1” for the dependent variable